Solar-Integrated Battery Energy Management for Intelligent Greenhouse Climate Automation

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Abstract

- This project develops a solar-powered battery-based energy system to operate an automated greenhouse climate control unit.
- A microcontroller-based control strategy regulates temperature and humidity using real-time sensor data for sustainable agriculture.

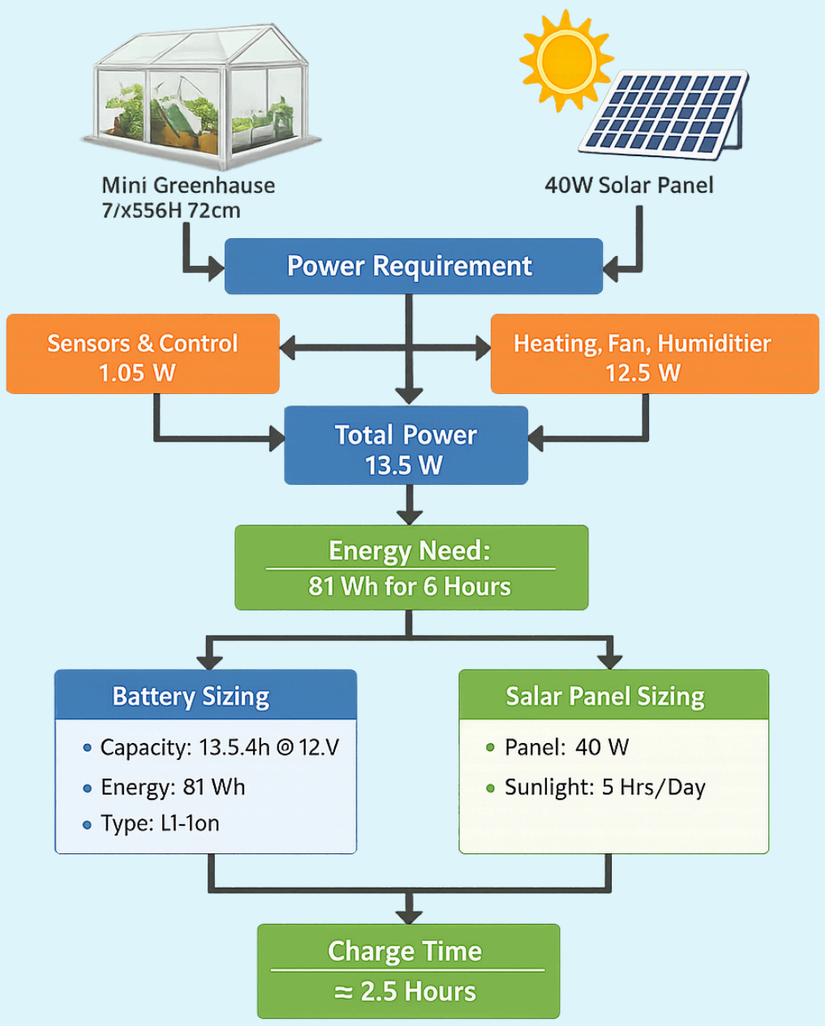
Objectives

- Design a solar-powered battery charging and management system for greenhouse applications.
- Implement automatic temperature and humidity control using sensors and a microcontroller.
- Reduce grid dependency and energy consumption through renewable energy utilization.

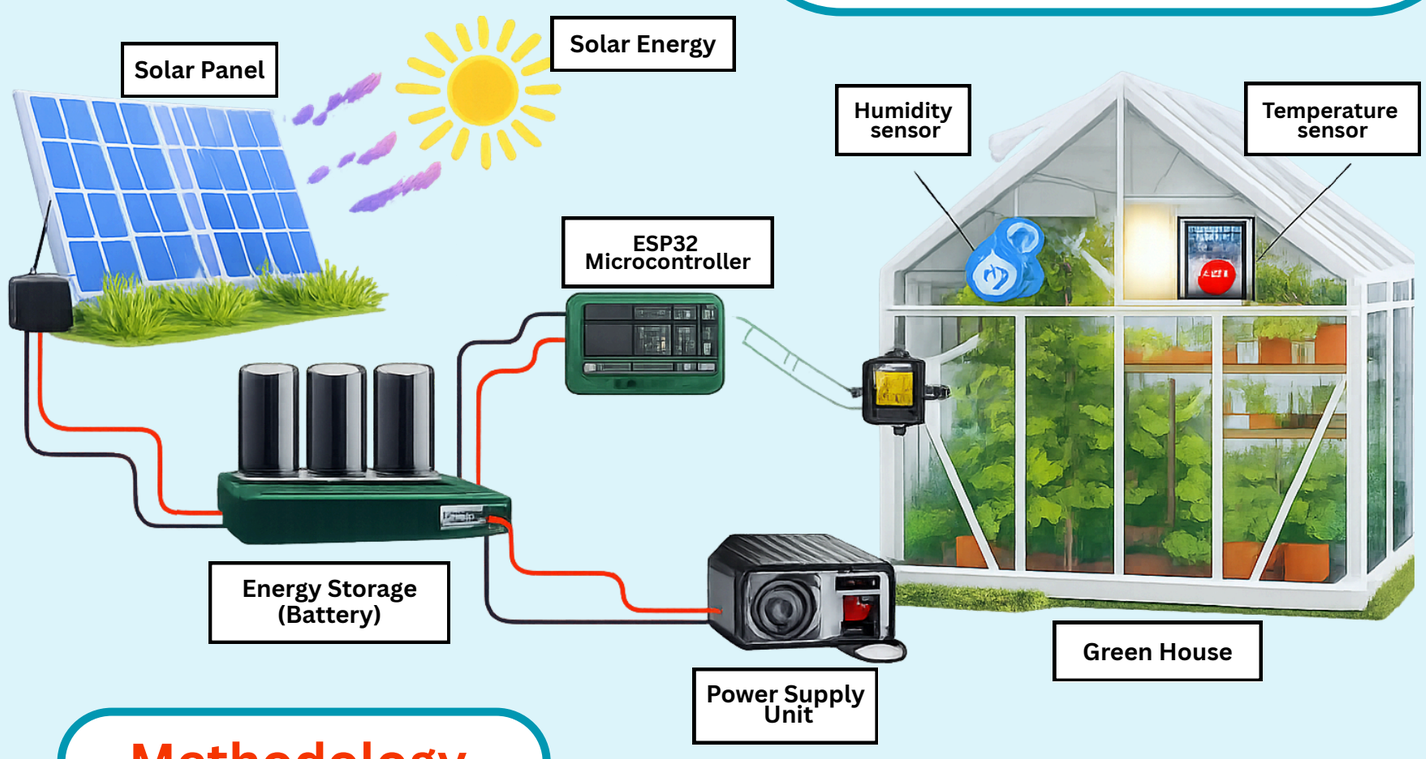
Used Materials

Solar Panel (6V, 1.1W, 200mA)
Rechargeable Battery (Li-ion / Lead-acid, 6–12V)
Battery Charge Controller
DC–DC Converter
Microcontroller (ESP32 Dev Module)
Temperature & Humidity Sensor (DHT22)
Cooling Fan
Humidifier
Relay Module
Logic Level Shifter

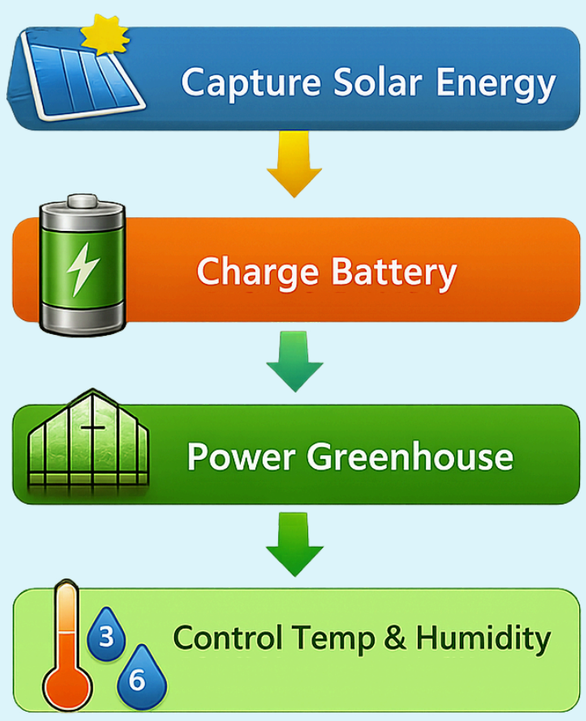
Power Estimation



Diagram



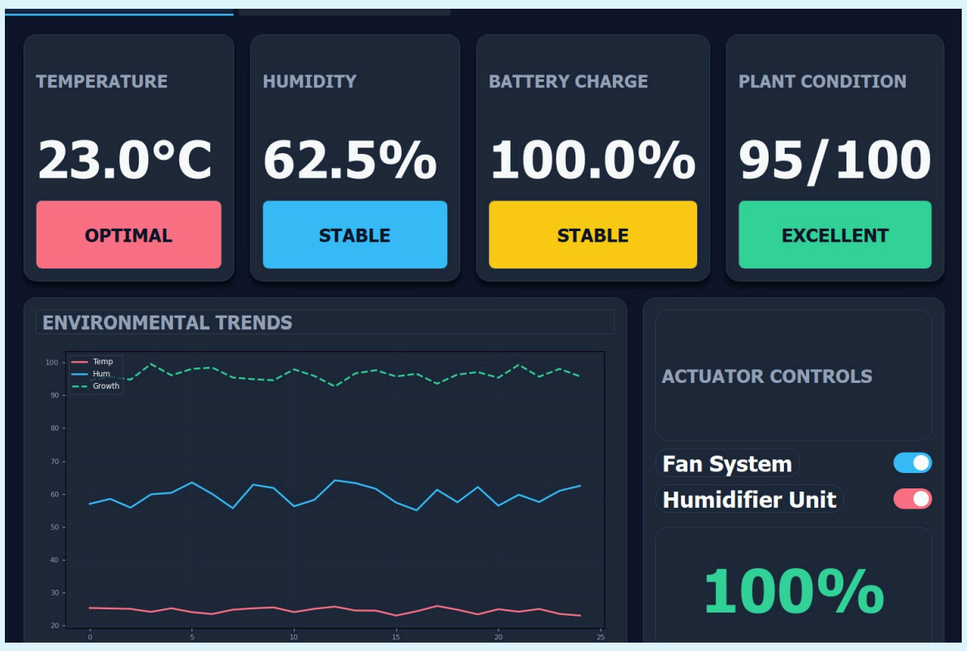
Methodology



Results

- Stable greenhouse temperature maintained within the desired range
- Automatic humidity regulation with minimal fluctuation
- Reliable energy supply ensured through battery storage
- Improved system stability during cloudy conditions and nighttime

User Interface



Conclusion

- The proposed system successfully demonstrates a solar-powered battery-based greenhouse climate automation system.
- Battery energy storage provides reliable, continuous operation compared to short-term storage devices.
- The system is energy-efficient, environmentally friendly, and suitable for smart agricultural applications.

Limitation

- Battery lifespan depends on charge–discharge cycles
- Initial battery cost and replacement over time
- Performance varies with solar irradiation availability

Future Scope

- IoT-based remote monitoring and mobile application control
- AI-based climate prediction and optimization
- Scaling for large commercial greenhouse installations